

WEEK 2 – POWER BI TASK

Data Modeling, DAX & Sales Dashboard

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Course: B.Sc Data Science and Artificial
Intelligence

Introduction

This week focuses on developing an interactive Superstore Sales Dashboard using Power BI. The task involves establishing a structured data model, creating foundational DAX measures, and designing interactive visualizations to monitor sales, profit, orders, customers, quantity, and regional performance. The dashboard will use KPI cards, charts, slicers, cross-filtering, tooltips, drill-down functionality, and a decomposition tree to support interactive analysis.

Figure 1: Initial Superstore Sales data model before establishing the star schema

Table: samplesuperstore (10,194 rows)

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Region
331	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	West
332	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	West
369	US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049	West
370	US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049	West
373	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West
375	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West
377	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West
723	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	West
727	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	West
1409	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West
1410	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West
1411	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West
1422	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1427	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1428	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1430	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1431	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1432	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West
1446	US-2023-1068305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	West
1449	US-2023-1068305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	West
1468	US-2023-153619	03 November 2023	03 November 2023	Same Day	TC-21535	Tracy Collins	Home Office	United States	Los Angeles	California	90049	West
1816	US-2023-138737	07 December 2023	10 December 2023	First Class	FP-14320	Frank Preis	Consumer	United States	Los Angeles	California	90049	West
1859	US-2023-113579	13 December 2023	15 December 2023	Second Class	KD-16345	Katherine Duchi	Consumer	United States	Los Angeles	California	90049	West
1886	US-2023-136399	16 December 2023	17 December 2023	First Class	CC-12100	Chad Cunningham	Home Office	United States	Los Angeles	California	90049	West
1891	US-2023-136399	16 December 2023	17 December 2023	First Class	CC-12100	Chad Cunningham	Home Office	United States	Los Angeles	California	90049	West

Modeling → New table

DimCustomer

DimProduct

DimDate

DimRegion

Figure 2: Star schema data model showing fact and dimension tables with established relationships

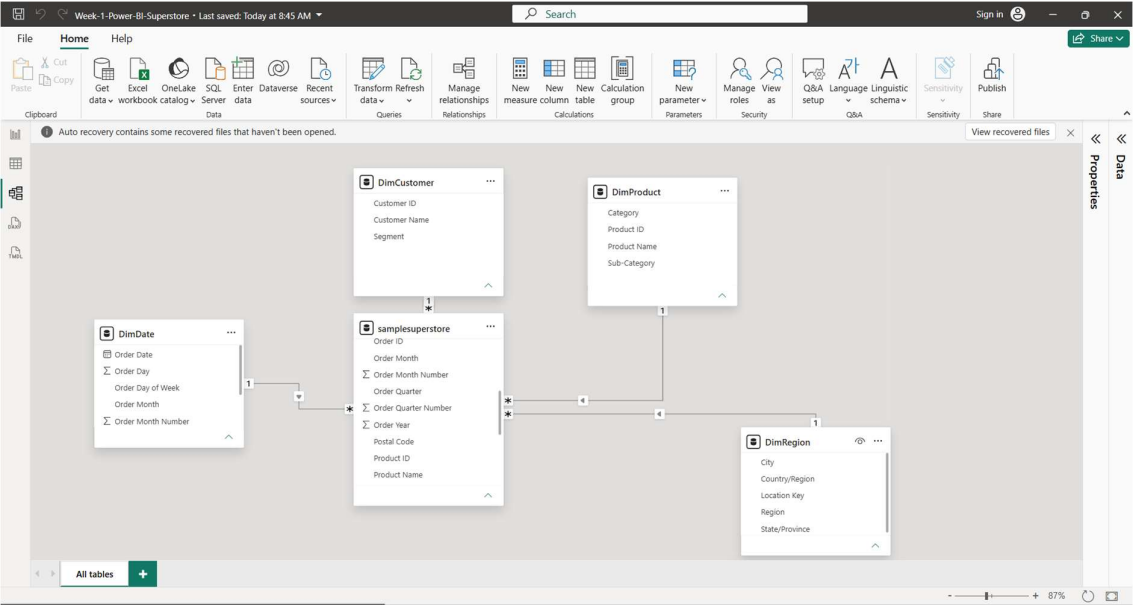


Figure 3: Dedicated _Measures table created for centralized DAX measures

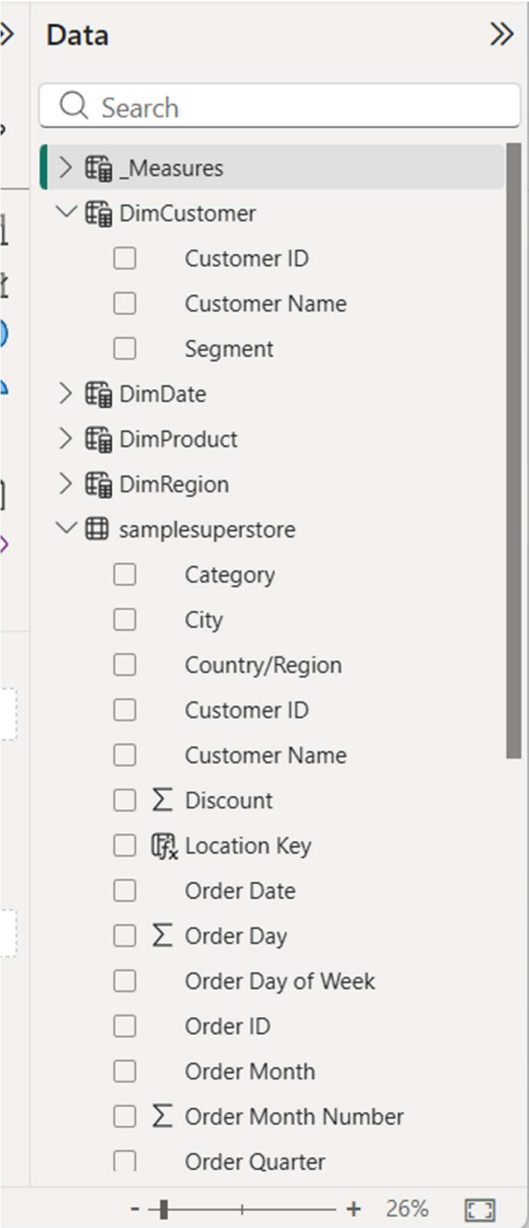


Figure 4: Foundational DAX measures created for sales, profit, orders, quantity, average sales, and profit margin analysis

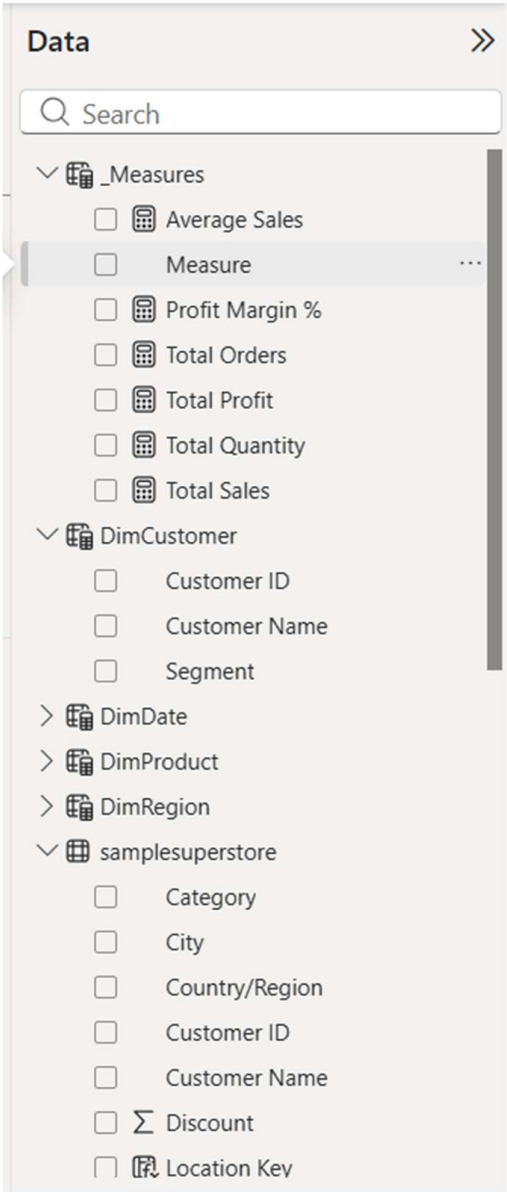


Figure 5: Executive summary displaying key sales, profit, order, quantity, and margin KPIs

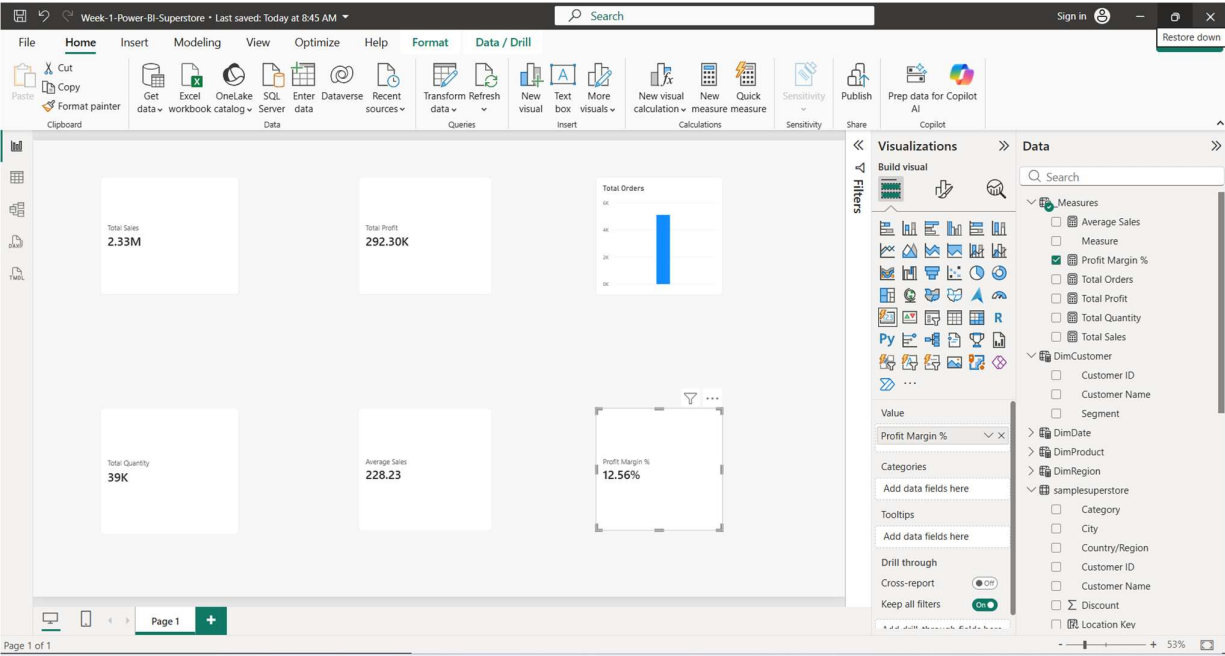


Figure 6: Category-wise sales and profit performance analysis

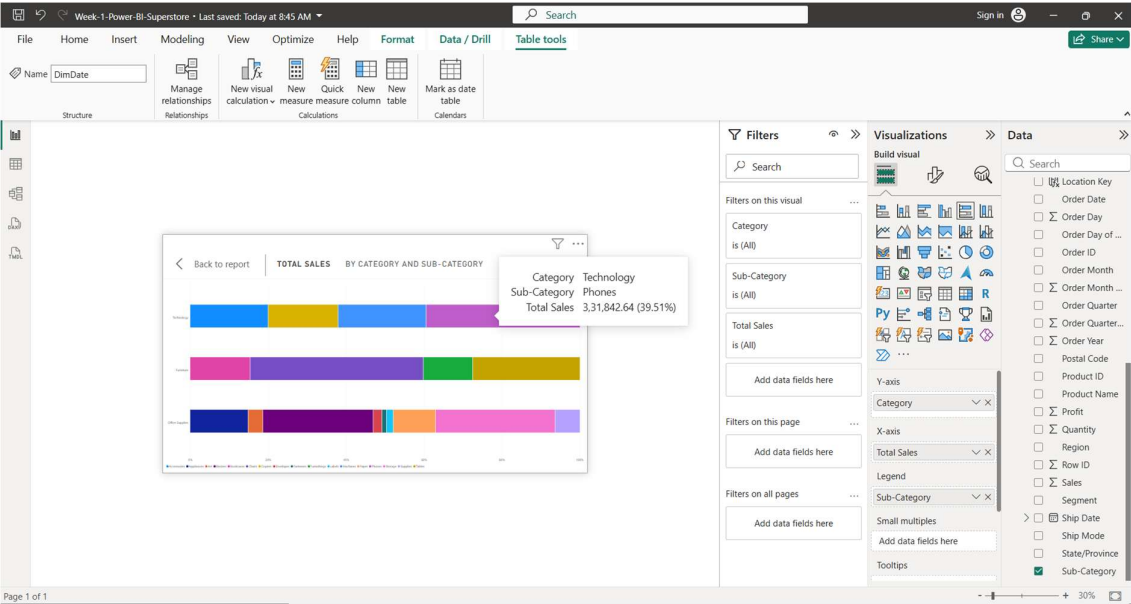


Figure 7: Regional profitability breakdown across Superstore sales regions

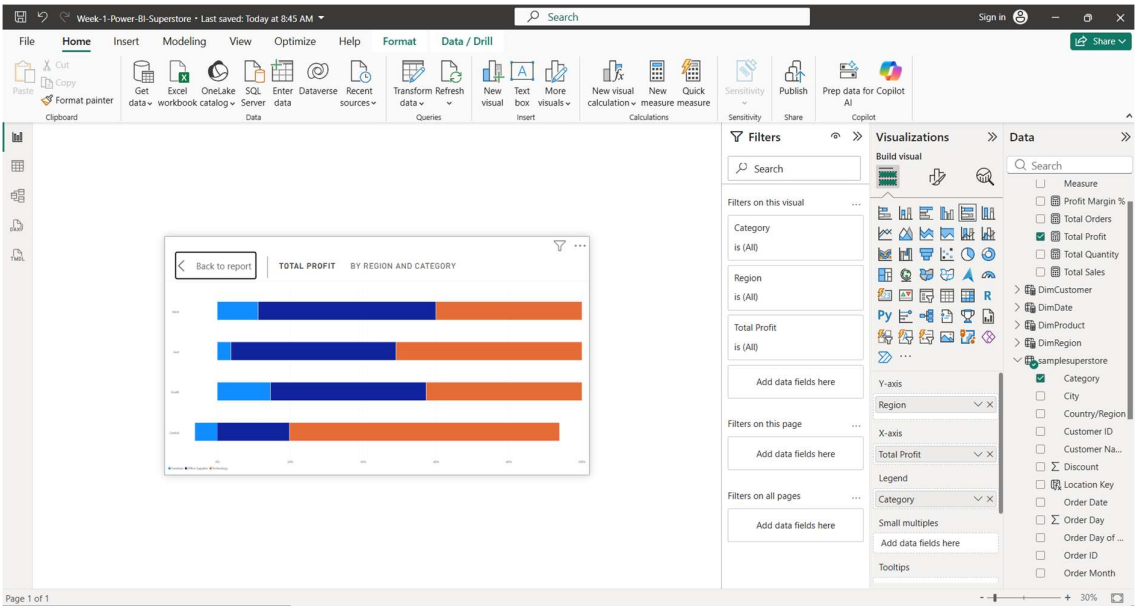


Figure 8: Sales trend over time using the Superstore date hierarchy

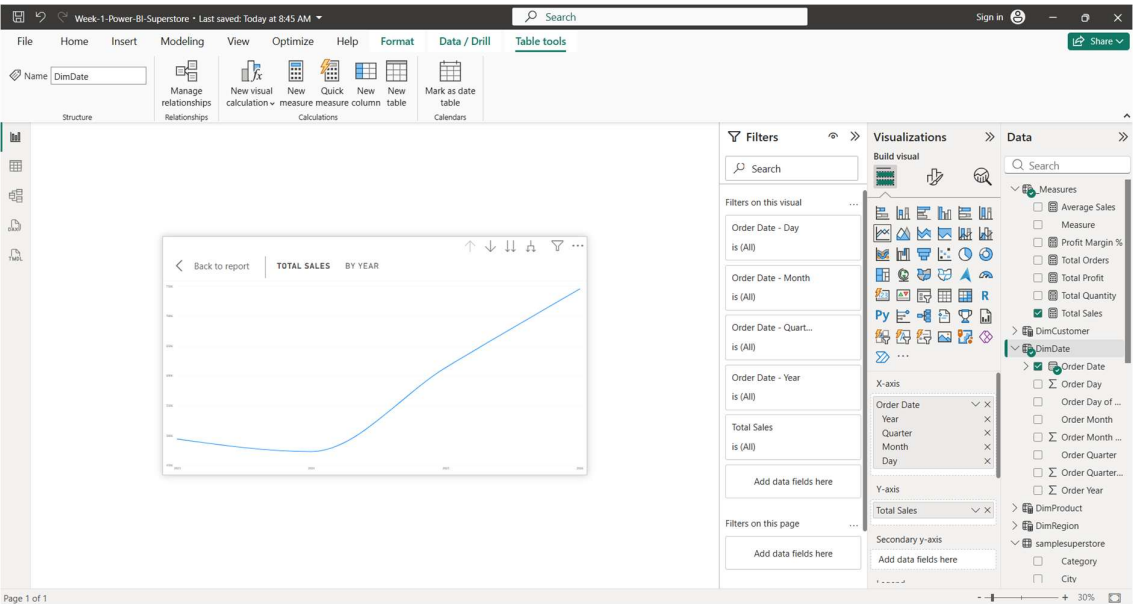


Figure 9: Decomposition Tree providing interactive analysis of sales across multiple dimensions

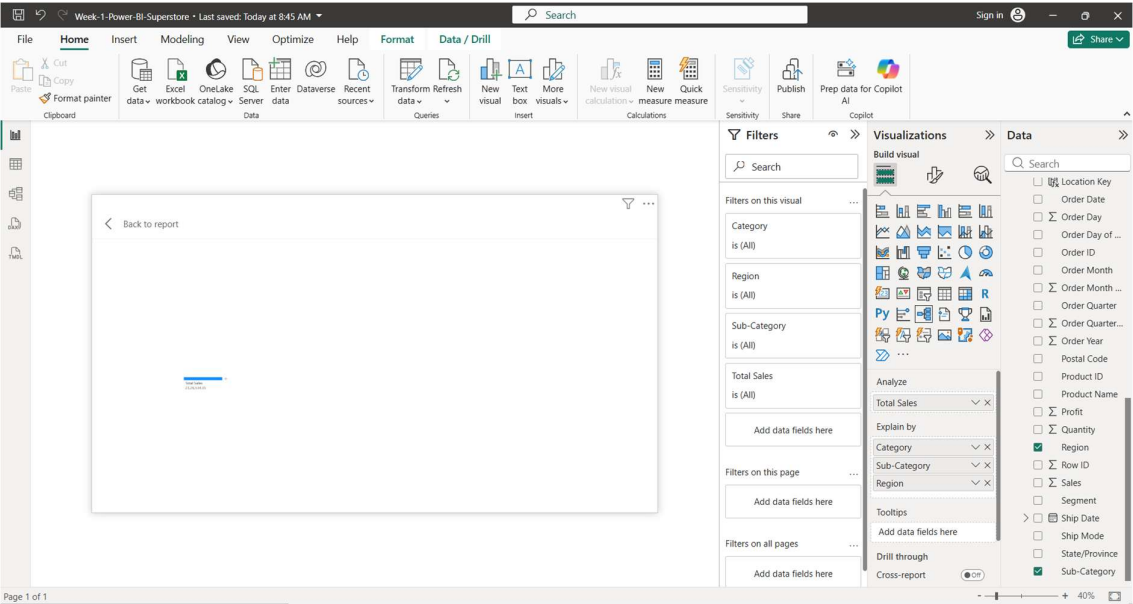


Figure 10: Interactive slicers configured for filtering dashboard results by key business dimensions

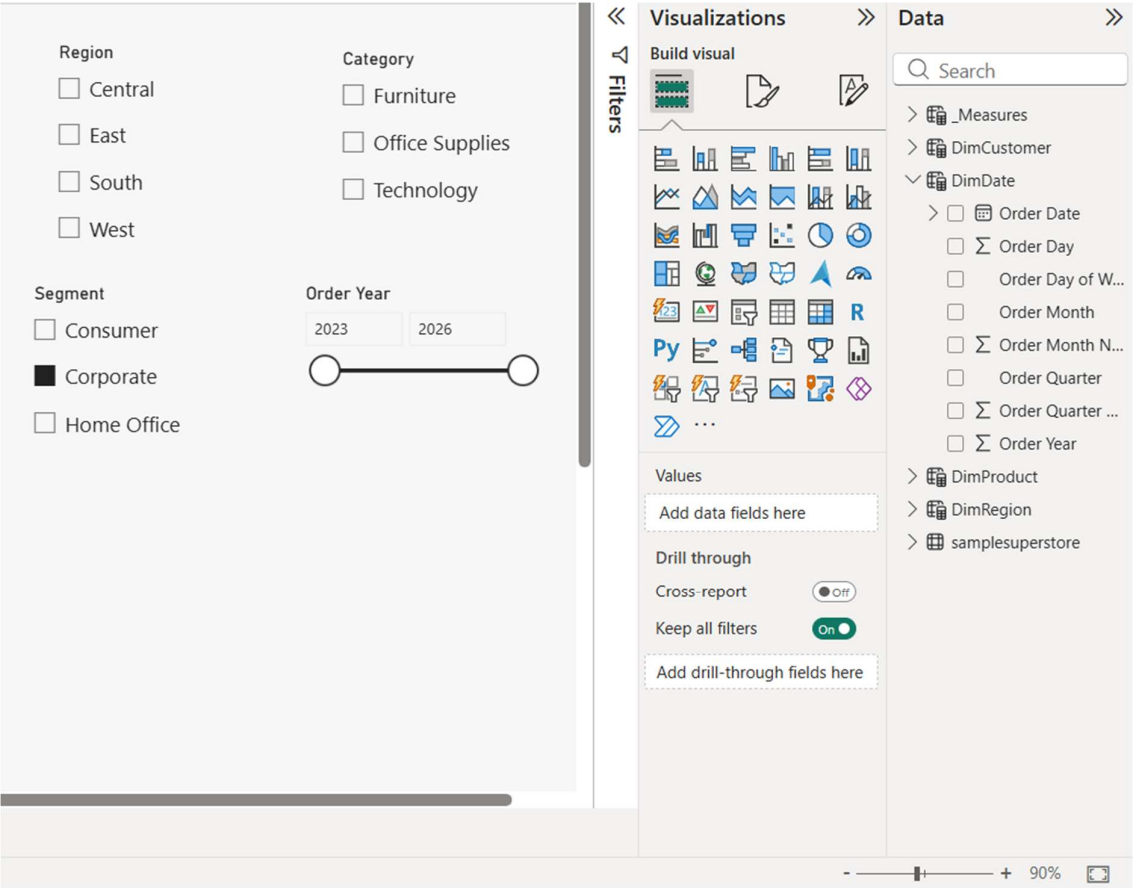


Figure 11: Cross-filtering interaction between dashboard visuals for dynamic analysis

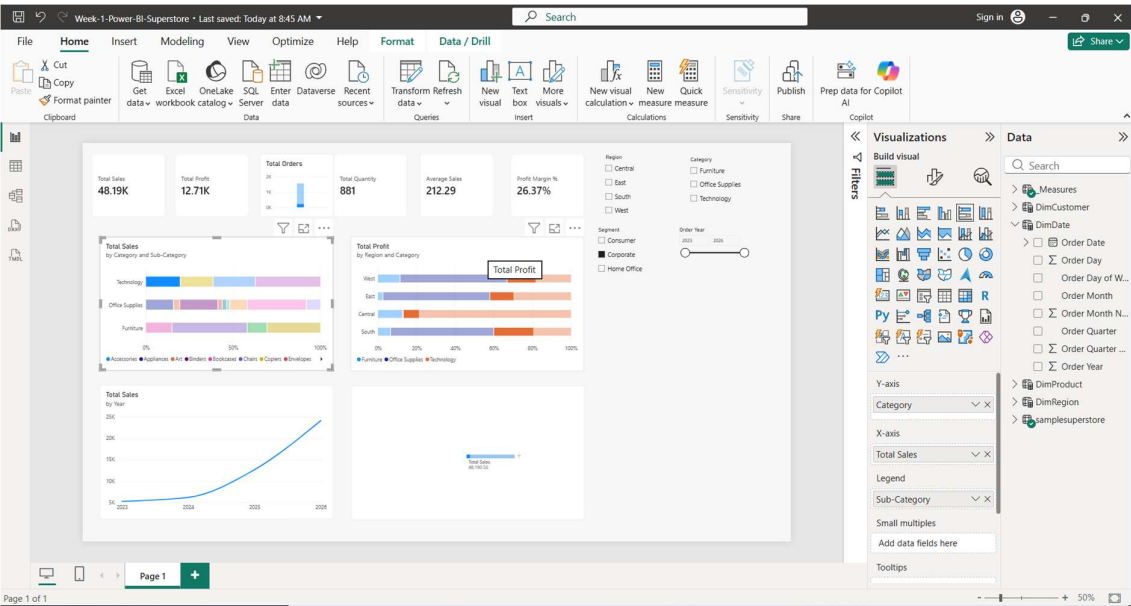


Figure 12: Drill-down functionality configured for detailed time-based sales analysis

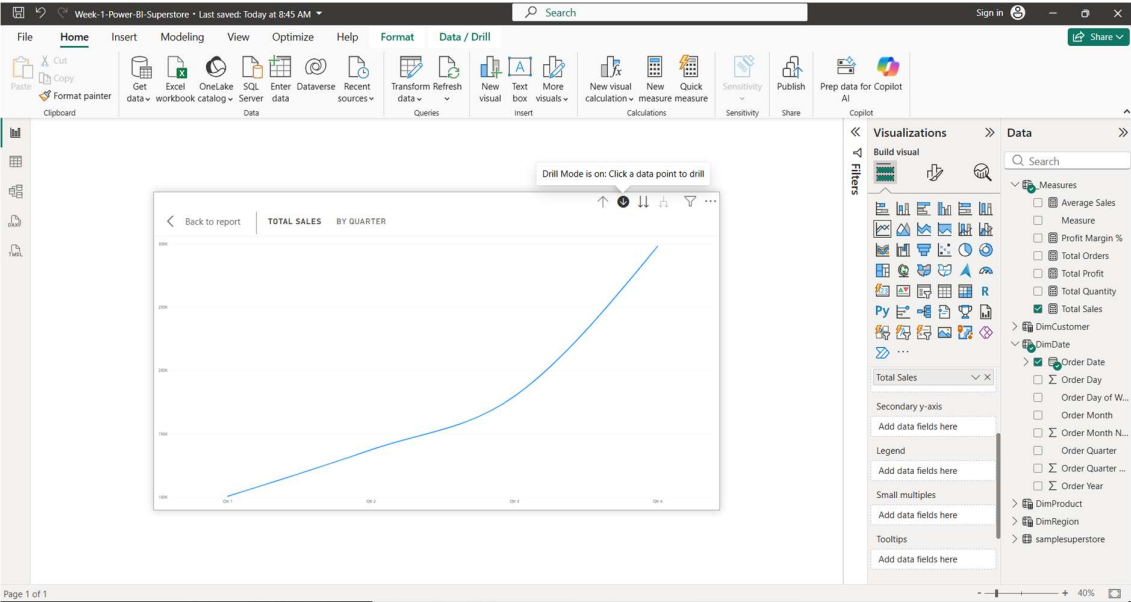
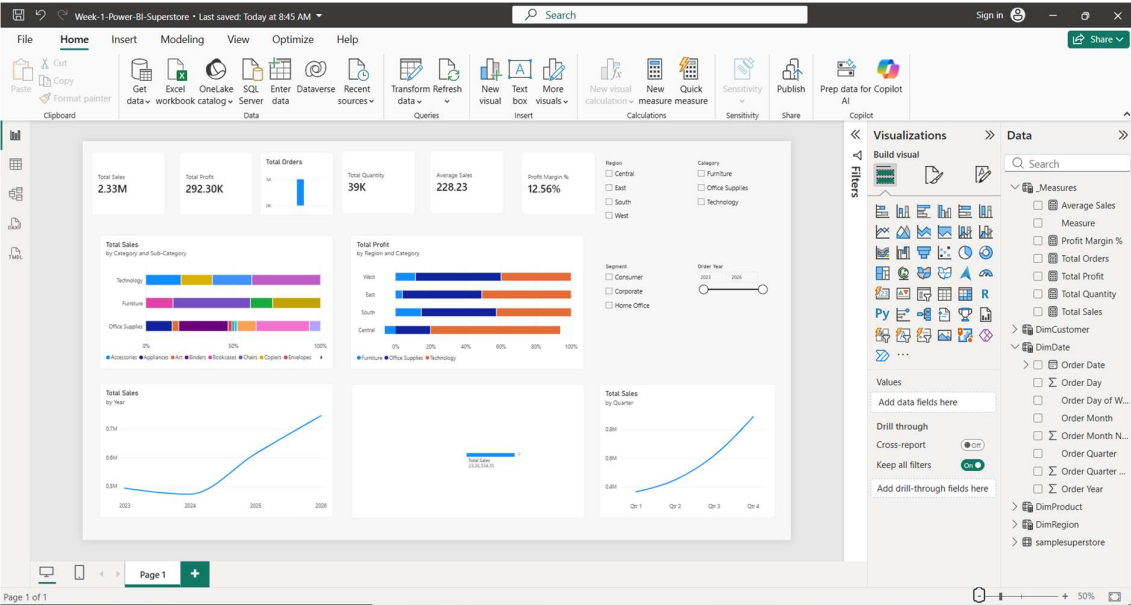


Figure 13: Final interactive Superstore Sales Dashboard integrating KPIs, visual analysis, slicers, and interactive filtering



Conclusion

The Superstore Sales dataset was successfully transformed into a structured star schema by establishing relationships between the fact and dimension tables. A dedicated _Measures table was created with foundational DAX measures for Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin %. An interactive dashboard was developed using KPI cards, category and regional analysis charts, a sales trend line chart, and a decomposition tree. Slicers, cross-filtering, and drill-down functionality were also configured to support interactive business analysis. The completed dashboard provides management with an integrated view of sales, profit, orders, quantity, and regional performance.